

Immersive Graphics in Education: A VR Rat Dissection Case Study Framed by Three Essential Abilities

Aftab Amin*

The Hong Kong University of Science and Technology
Hong Kong, Hong Kong
aftabamin@ust.hk

Ka Man Yan*

The University of Hong Kong
Hong Kong, Hong Kong
aidenyan@hku.hk

Abstract

This paper presents an immersive VR pedagogy project developed at HKUST to support undergraduate life science students in learning rat anatomy and dissection. Traditional anatomy teaching frequently encounters ethical concerns, substantial material costs, and student anxiety when handling specimens. Our VR platform provides a safe, repeatable, and interactive environment where students can explore anatomical structures, practice procedures, and build confidence prior to wet lab engagement. Baseline surveys revealed that 64.6% of students reported increased preparedness following VR practice, while 79.3% expressed satisfaction with the learning experience, citing reduced anxiety and enhanced spatial understanding as primary benefits. Beyond this case study, we position the project within broader educational futures through three essential dimensions. First, immersive HCI can foster empathetic understanding among peers while enabling students to develop self-efficacy and emotional resilience. Second, the integration of AI and multimodal tools enables students to engage in creative expression, combining scientific learning with narrative construction and cross-disciplinary perspectives. Finally, as educational technologies increasingly collect and process student information, cultivating critical thinking about educational technology becomes essential for responsible digital citizenship. By situating a concrete VR teaching practice within these three educational dimensions—empathetic engagement, creative expression, and critical evaluation—this contribution demonstrates not only the practical value of immersive tools in STEM education but also the importance of aligning technological innovation with human-centered pedagogical values.

CCS Concepts

• **Applied computing** → Education; Interactive learning environments; Education; E-learning; • **Computing Methodologies** → Modeling and simulation; simulation types and techniques; Interactive simulation.

Keywords

Virtual Reality, Gamification, Educational Technology, Immersive Learning, Anatomy Education, Active Learning

*Corresponding author.

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1 Introduction

This paper details the development and implementation of an innovative virtual reality (VR) pedagogy project, serving as a compelling case study to illustrate the profound potential of advanced graphics technologies within higher education. The core focus of this initiative is the instruction of rat anatomy and dissection, a subject traditionally delivered using physical specimens [Javvaji et al., 2024; Neyem et al., 2025]. This conventional approach, however, presents a multitude of challenges, including significant ethical considerations [de Villiers, 2011], substantial material costs [Stimec et al., 2010], and a documented propensity to induce high levels of anxiety among students [Ma et al., 2025].

Our meticulously designed VR platform offers a transformative alternative: a safe, entirely repeatable, and highly interactive learning environment [Lewis et al., 2024]. Within this digital space, students are empowered to repeatedly practice complex dissection procedures, meticulously explore intricate anatomical structures in a dynamic 3D medium, and systematically build confidence. This invaluable preparation occurs prior to their eventual transition into a traditional wet lab setting, thereby mitigating initial apprehension and enhancing practical proficiency [Withers et al., 2024].

The practical application of this VR pedagogy is currently integrated into the LIFS1904 course at HKUST (Hong Kong University of Science and Technology) [Center for Education Innovation, 2024; Center for Education Innovation, 2025]. This foundational course introduces approximately 300 undergraduate students annually to a broad spectrum of fundamental laboratory practices. The curriculum encompasses four distinct experimental modules: rat dissection, fruit fly genetics, forensic DNA fingerprinting, and protein biochemistry. Among these diverse modules, the rat dissection experiment has been consistently identified as a particularly significant source of stress for students [Zhang et al., 2021], manifesting in both psychological distress and considerable logistical complexities for instructors and lab assistants. The VR platform directly addresses these long-standing issues, offering a scalable and student-centric solution to a critical pedagogical challenge.

2 Addressing Pedagogical Challenges Through Virtual Reality

The development of a Virtual Reality (VR) application for rat dissection directly addresses critical ethical concerns and practical resource limitations frequently encountered in traditional biological education [Dodevska et al., 2025]. This innovative approach offers a transformative solution to several long-standing challenges:

Substituting physical specimens: The VR platform offers a safe, repeatable, and highly interactive learning environment, significantly reducing or even potentially eliminating the need for animal cadavers [Javvaji et al., 2024]. This directly tackles the ethical dilemmas associated with animal use in education, an issue that not only raises significant moral concerns but also involves substantial material costs for procurement, preservation, and disposal of specimens [Stimec et al., 2010]. By providing a realistic virtual alternative, the VR app aligns with the principles of the "3 Rs" (Replacement, Reduction, Refinement) in animal research and education [de Villiers, 2011].

Safe practice environment: This VR application provides an invaluable opportunity for students to practice complex dissection procedures and freely make mistakes in a virtual space without causing harm to actual animals [Lewis et al., 2024]. This capability is particularly beneficial for pre-lab preparation, allowing students to familiarize themselves with anatomical structures, instrument handling, and procedural steps before engaging with any physical specimens, if they are still used. Moreover, for the actual dissection process, the VR environment facilitates an important learning journey where experimentation and error are encouraged as part of the educational process. Students are encouraged to treat these virtual lives with the same respect as real ones, fostering a sense of responsibility and reverence for life, while ensuring that the fear of making mistakes does not impede their learning progress or exploration within the simulation [Withers et al., 2024]. This approach promotes a deeper understanding of anatomy and surgical techniques through repeated, risk-free practice.

2.1 Addressing Resource Limitations in Laboratory Education Through Virtual Reality

The integration of a Virtual Reality (VR) application offers a comprehensive solution to several persistent resource limitations faced in traditional laboratory education, particularly in disciplines requiring hands-on experience with biological specimens [Dodevska et al., 2025]. This innovative approach not only mitigates existing challenges but also enhances the learning experience for students.

2.1.1 Cadaver Availability. A significant hurdle in biological and anatomical studies is the inconsistent availability of cadavers, such as rat cadavers for dissection exercises. Events like global pandemics (e.g., COVID-19) can severely disrupt supply chains, making it challenging to acquire these essential specimens [Ma et al., 2025]. Furthermore, even in normal circumstances, procuring the ideal quantity and desired gender ratio (male and female rats) for all lab sessions can be difficult due to ethical considerations, regulations, and logistical complexities [Zhang et al., 2021]. The VR application effectively circumvents this issue by providing highly detailed and

interactive 3D virtual models of rat anatomy. This ensures that every student has access to a consistent and readily available "specimen" for virtual dissection, regardless of external supply fluctuations.

2.1.2 Increased Group Sizes and Equitable Access. As student enrolment in science programs continues to rise, laboratory resources often struggle to keep pace, leading to increased group sizes for practical sessions [Lewis et al., 2024]. When faced with limited cadaver availability, this often means that students within larger groups may have reduced individual dissection time or limited exposure to different specimen types. The VR application addresses this by allowing students to efficiently examine both male and female 3D rat models in a virtual environment. This is particularly beneficial even if students can only physically dissect one gender due to resource constraints. The VR platform ensures that all students gain a comprehensive understanding of anatomical differences, promoting equitable learning opportunities despite physical lab limitations [Dodevska et al., 2025].

2.1.3 Manpower and Space Shortages. Traditional laboratory settings are inherently limited by physical space and the availability of teaching assistants or instructors to supervise practical sessions [Javvaji et al., 2024]. Growing student numbers often exacerbate these issues, leading to crowded labs and strained supervision. The VR platform offers a transformative solution by allowing multiple students to engage in virtual laboratory exercises simultaneously within a virtual classroom. This innovative approach frees up valuable physical bench space in the teaching laboratories, as students can utilize the VR platform independently or in smaller, more flexible arrangements. Moreover, it allows students to productively use their time by engaging with the VR platform while waiting for their turn in the practical lab, optimizing learning efficiency and reducing idle time [Zhang et al., 2021]. This directly helps resolve critical issues related to manpower and space shortages, creating a more efficient and less constrained learning environment.

2.1.4 Cost-Efficiency and Long-Term Sustainability. While the initial development of a custom VR platform might represent an upfront investment, it offers substantial long-term cost efficiencies compared to relying on commercial Virtual Learning Systems (VLS) or other proprietary VR solutions [Stimec et al., 2010]. Developing an in-house platform avoids excessive recurring licensing costs, which can accumulate significantly over time and become a substantial burden on educational budgets. This approach promotes financial sustainability for the institution and allows for greater control over the platform's features, updates, and scalability. For example, it can also help establish the fundamentals of human anatomy development for the next medical school potentially developed.

2.1.5 Reduced Reagent and Time Waste, Enhanced Preparedness. Beyond anatomical dissections, the VR application extends its utility to other related lab exercises, such as DNA fingerprinting and in vitro digestion experiments [Neyem et al., 2025]. These procedures often involve expensive reagents and require precise execution to yield accurate results. The VR app provides a safe and consequence-free environment for students to practice these complex procedures repeatedly. By rehearsing steps and identifying potential pitfalls in the virtual space, students can minimize errors when they transition

to the physical lab. This proactive approach significantly reduces the waste of costly reagents and valuable lab time, ultimately leading to more successful and efficient practical sessions. The VR platform essentially serves as a preparatory tool, enhancing student confidence and competence before engaging with real-world laboratory challenges.

3 Case Study: HKUST VR Pedagogy Project – Advancing Experiential Learning Through Virtual Reality

The Hong Kong University of Science and Technology (HKUST) has been at the forefront of integrating cutting-edge technology into its pedagogical approaches, exemplified by the ambitious HKUST VR Pedagogy Project [Division of Life Science, 2025]. This initiative, conceived under the robust TLIP (Teaching and Learning Innovation Project) framework, sought to revolutionize experiential learning in scientific disciplines by leveraging the immersive capabilities of virtual reality.

Phase 1: Pioneering Virtual Dissection

The initial phase of the project concentrated on addressing a critical need in life science education: practical laboratory skills. Phase 1's flagship development was a sophisticated rat dissection module [Center for Education Innovation, 2024]. This module was meticulously built using Unity, a powerful real-time 3D development platform, ensuring high fidelity graphics and interactive capabilities. It was designed for deployment on Oculus Quest 2/3 headsets, providing students with accessible and intuitive access to the virtual environment (Fig 1. A-E and Fig. 2).

The core objective of this module was to simulate the intricate anatomy lab environment. This virtual simulation allowed students to engage in hands-on dissection practice in a risk-free setting. By practicing virtually before entering the physical wet lab, students could familiarize themselves with the dissection procedures, anatomical structures, and instrument handling, thereby enhancing their confidence and preparedness, and ultimately optimizing their learning outcomes in the traditional laboratory setting. This pre-lab virtual practice also significantly reduced the demand for physical specimens, aligning with ethical considerations and sustainability goals.

Phase 2: Expanding Horizons with Narrative Gamification

Building upon the demonstrable success and positive feedback from Phase 1, Phase 2 of the HKUST VR Pedagogy Project significantly broadened its scope [Center for Education Innovation, 2025]. This expansion saw the development of additional VR modules catering to a wider array of life science experiments (Fig 1. F-G and Fig 3. A-C). These included:

- **Genetics (using fruit flies):** This module aimed to provide students with a dynamic and interactive way to explore genetic principles, such as inheritance patterns and genetic crosses, through virtual manipulation of fruit flies.
- **Forensic DNA Fingerprinting:** Stepping into the realm of applied science, this module immersed students in the process of DNA analysis, crucial for forensic investigations.

- **Protein Biochemistry:** This module offered a novel way to visualize complex molecular structures and biochemical reactions, providing a deeper understanding of protein functions and interactions.

A pivotal innovation in Phase 2 was the incorporation of a narrative gamification approach. This strategic design choice aimed to elevate student engagement and situational immersion beyond traditional simulations. For instance, the forensic DNA fingerprinting module was ingeniously framed within a compelling crime scene investigation scenario (Fig 1. F-G). This narrative context transformed what could have been a dry technical exercise into an exciting problem-solving challenge, significantly boosting student motivation and curiosity.

Gamification Design Elements: Enhancing Engagement and Knowledge Retention

The gamification elements were not merely superficial additions but were meticulously integrated to enhance both engagement and knowledge retention. Key design features included:

Task Unlocking: This mechanism progressively revealed new challenges and information as students mastered earlier concepts, fostering a sense of achievement and encouraging continuous learning (Fig 1. A-B).

Deliberate Error-Based Learning: A particularly innovative feature was the simulation of consequences for incorrect actions. For example, in the dissection modules, performing a dissection incorrectly might result in virtual bleeding, providing immediate, tangible feedback on errors. This "learning by doing and correcting mistakes" approach, without the ethical or material costs of real-world errors, significantly lowered student anxiety and encouraged bolder experimentation and critical self-correction.

Interactive Assessments: Unlike traditional assessments, these were seamlessly embedded directly within the VR environment. This allowed for real-time evaluation of student understanding and application of concepts, providing immediate feedback and reinforcing learning.

Student Feedback and Impact: A Transformative Learning Experience

The feedback from students who engaged with the VR modules was overwhelmingly positive and underscored the transformative impact of this pedagogical innovation. Students consistently reported that the VR modules offered:

Increased Flexibility: The ability to access and practice the experiments at their own pace and convenience, outside of fixed lab schedules.

Enhanced Safety: A risk-free environment where complex or hazardous procedures could be practiced without any personal or material danger.

Improved Repeatability: The modules allowed for unlimited practice repetitions, enabling students to refine their skills and consolidate their understanding without resource constraints.

Crucially, students highlighted that the ability to make mistakes without ethical repercussions or material costs significantly lowered their anxiety levels. This freedom from fear of failure encouraged active exploration and experimentation within the virtual environment, leading to a deeper and more meaningful learning experience. Furthermore, the immersive 3D graphics provided a powerful tool

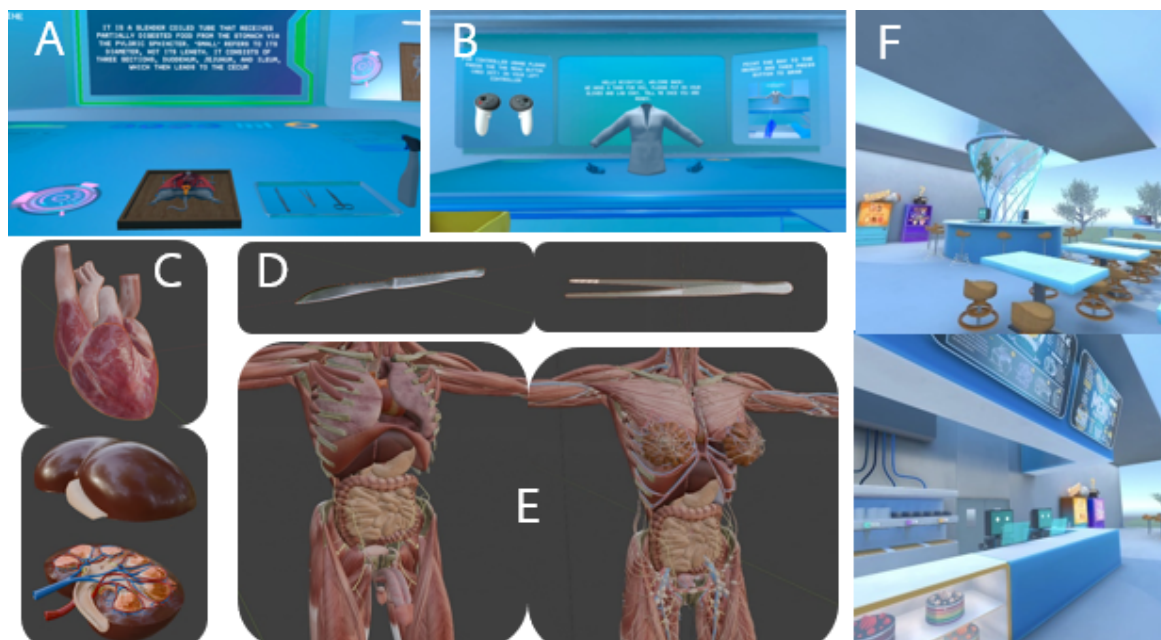


Figure 1: Phase 1 and Phase 2 Virtual Reality Pedagogy Assets. (A) Rat dissection and organ analysis gamification, (B) Health and safety gamification, (C) Human organs, (D) Dissection tools, (E) Male and female human anatomy, (F) Phase 2 Crime scene DNA sample collection environment. Representative images are shown.

for comprehending complex spatial anatomy, offering a level of visualization and interaction that traditional two-dimensional learning materials could not match. The HKUST VR Pedagogy Project stands as a testament to the power of virtual reality in creating engaging, effective, and ethically sound learning environments for the next generation of scientists.

4 Methodology and Implementation

The comprehensive implementation of Virtual Reality Pedagogy (VRP) within the curriculum was meticulously orchestrated through a multi-stage teaching workflow, strategically designed to cultivate superior student preparedness and bolster confidence prior to their engagement in physical laboratory experiments. This workflow was initiated with a critical pre-lab preparation phase, where students immersed themselves in interactive VR practice modules. These modules were not merely simulations but carefully constructed digital environments that allowed students to thoroughly familiarize themselves with complex experimental procedures, deepen their understanding of underlying scientific concepts, and even anticipate potential challenges, thereby demystifying the subsequent hands-on experience (Fig. 4).

Following this foundational virtual preparation, students transitioned seamlessly to the hands-on wet lab experiment. This crucial stage provided the practical arena for students to directly apply the theoretical knowledge and procedural skills acquired in the VR environment to a real-world setting, manipulating actual equipment and observing authentic chemical or biological reactions. The learning cycle was then brought to a comprehensive conclusion with rigorous assessments. These evaluations were designed to be

multifaceted, examining not only students' theoretical understanding of the subject matter but also their practical application skills, critical thinking, and problem-solving abilities within the context of the experiments. This integrated pedagogical design consistently demonstrated that students entered the physical laboratory environment not only significantly better prepared for the practical demands but also imbued with a demonstrably higher level of confidence in their own capabilities, leading to more effective and less anxiety-ridden learning experiences.

To ensure the continuous efficacy and ongoing improvement of the VRP intervention, a robust action research methodology was systematically employed. This iterative and profoundly reflective approach facilitated an ongoing, dynamic assessment and refinement of the pedagogical design. Data collection for this research was intentionally multifaceted and comprehensive, incorporating a variety of qualitative and quantitative methods. These included detailed surveys administered to a broad student population, structured focus groups that allowed for deeper exploration of shared experiences and perceptions, and in-depth one-on-one interviews with both students and teaching staff. These diverse methods were instrumental in gathering rich, nuanced insights from all key stakeholders, thereby providing a truly holistic perspective on various critical aspects of the VRP. Key areas of investigation included the usability and accessibility of the VR system, its measurable impact on specific learning outcomes (such as procedural accuracy, conceptual understanding, and retention), and the overall engagement levels of participants within the VR learning environment. The actionable findings derived from this rigorous research directly informed and guided subsequent iterations and strategic

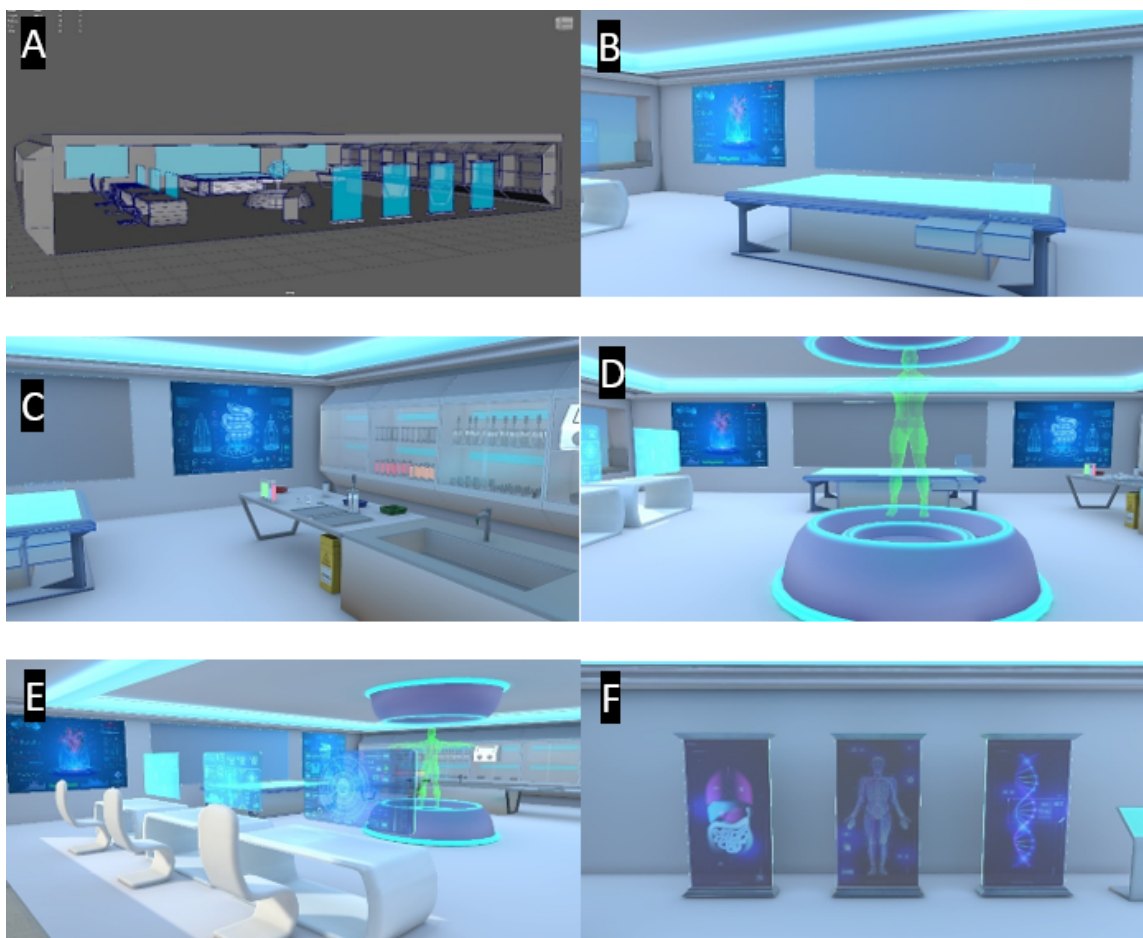


Figure 2: VR Environment. (A) Floor Plan of the VR Classroom, (B-E) VR environment assets, (F) Exercise Holograms, Anatomy, DNA, Biomolecule Digestion. Representative Images are shown.

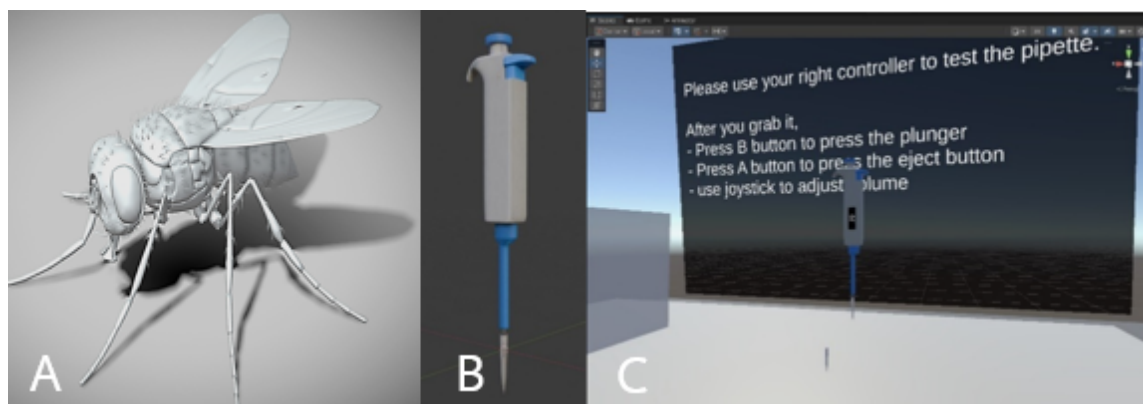


Figure 3: Phase 2 Virtual Reality Pedagogy Assets. (A) Fly, (B) Pipette, (C) Pipetting instructions. Representative images are shown.



Figure 4: Trained Teaching Assistants teaching UG students using virtual reality in the LIFS1904 Laboratory.

improvements to the VRP, ensuring its continuous evolution and optimization.

A pivotal element underpinning the remarkable success and sustained deployment of the VRP was the proactive and comprehensive training provided to both teaching assistants (TAs) and research assistants (RAs). This intensive training program was meticulously structured to equip them with two critical sets of skills. Firstly, it focused on imparting the necessary technical proficiency to expertly operate and troubleshoot the sophisticated VR systems, ensuring smooth and uninterrupted delivery of the VR modules. More importantly, the training emphasized effective pedagogical integration – guiding TAs and RAs on how to seamlessly incorporate these advanced technologies into their teaching practices, leveraging the unique affordances of VR to enhance instruction and student interaction. This strategic initiative served a dual and interconnected purpose: it directly supported the immediate operational needs of the project by ensuring competent in-field support, and concurrently, it contributed significantly to building a robust and sustainable capacity for educational innovation within the broader teaching team at the institution. By empowering TAs and RAs with these cutting-edge skills, the project effectively fostered a self-sustaining environment conducive to the future adoption and successful integration of advanced educational technologies across various disciplines.

The strategic allocation of resources was unequivocally a paramount factor in the exceptionally effective and widespread implementation of the VRP. The project was launched with a substantial initial deployment of over ten state-of-the-art Oculus headsets. This significant investment ensured broad accessibility for a large cohort of students, mitigating potential bottlenecks and maximizing participation. To facilitate the smooth and efficient operation of the VR modules and enable effective supervision, these high-tech headsets were thoughtfully complemented by an array of tablet PCs. These tablets empowered teaching staff to efficiently monitor student progress within the virtual environment in real-time, identify areas where students might be struggling, and provide immediate, targeted support or guidance as needed. This meticulously planned and integrated technological configuration was absolutely instrumental in enabling multiple students to concurrently engage with the VR modules within a single classroom setting. This simultaneous engagement capability maximized the pedagogical impact of the VRP, ensuring an immersive, interactive, and highly effective learning experience for a larger student cohort, ultimately scaling the benefits of virtual reality education.

5 Evaluation & Outcomes

Evaluation focused on three dimensions: learning outcomes, student motivation, and teaching efficiency.

Survey results demonstrated that students found VR helpful in preparing for the wet lab. 64.6% agreed that the VR experience increased their confidence and readiness, while 79.3% expressed overall satisfaction (Appendix A). Many students specifically noted that VR reduced their fear of handling a dead rat. One remarked, “It would emotionally prepare me for dissecting a real dead rat” (Appendix A), while another explained, “After the VR experience, I was more confident to do that and have some experience” (Appendix A).

Students also emphasized the value of visualizing anatomy in 3D. More than 80% agreed that the VR content was relevant to their learning, and 58.1% highlighted the usefulness of zooming and rotating models. As one student put it, “I could basically zoom in on the organs and understand what they look like, not just outside but inside as well” (Appendix A).

However, challenges were also noted. Around 30% of students reported physical or mental discomfort, including dizziness and headset strain, particularly for those wearing glasses. Others felt that the VR app should include more structured tasks, such as quizzes or guided missions, to avoid free exploration without clear learning objectives (Appendix A).

For teaching staff, VR proved beneficial by reducing the instructional burden. Students entered the wet lab with greater confidence and procedural understanding, requiring less remedial guidance from TAs. Resources such as specimens were used more efficiently, as students had already practiced in a risk-free VR environment.

6 Three Educational Dimensions

The VR pedagogy project not only addressed immediate challenges in teaching anatomy but also invited reflection on broader educational futures, particularly when situated within a framework of three essential abilities: psychological readiness, creative expression, and critical thinking [Mallek et al., 2024]. Together, these dimensions highlight how immersive graphics technologies can extend beyond cognitive learning outcomes to shape emotional, imaginative, and ethical aspects of education [Ma et al., 2025].

The first and central dimension is psychological readiness. Laboratory experiences such as rat dissection often provoke student anxiety, with several participants in the survey explicitly stating that they were “scared of dissecting a rat”. The VR intervention helped alleviate this barrier: 64.6% of students reported feeling more

confident and prepared for the wet lab after using the VR app, and 79.3% expressed overall satisfaction with the VR experience. One student reflected, “It would emotionally prepare me for dissecting a real dead rat”, while another noted, “After the VR experience, I was more confident to do that and have some experience”. These responses illustrate how immersive HCI can function as a form of emotional support, reducing stress and building confidence, thereby fostering a more inclusive and supportive learning environment [Dodevska et al., 2025].

A second dimension is creative expression. While the VR dissection module was designed primarily as a pre-lab tool, students consistently emphasized its value for exploration and visualization. More than 80% of respondents agreed that the information presented in the VR app was relevant to their learning outcomes, and 58.1% specifically endorsed the usefulness of interactive 3D features such as zooming and rotation [Javvaji et al., 2024]. One student commented, “I could basically zoom in on the organs and understand what they look like, not just outside but inside as well”. Others suggested further enhancements, including “adding quiz questions” or the ability “to annotate directly on the models”. Such feedback suggests that VR can evolve into a platform for creative engagement, allowing students not only to practice procedures but also to reframe and express knowledge in personalized ways [Ma et al., 2025]. When combined with AI-driven tools, VR could support multimodal forms of storytelling, narration, and annotation, blending scientific accuracy with imaginative interpretation [Stimec et al., 2010].

The third essential dimension is critical thinking. While the majority of students reported positive experiences, approximately 30% also described challenges related to the VR technology itself, including dizziness, motion sickness, and discomfort from wearing the headset, particularly among those who also wore glasses [Lewis et al., 2024]. Several students questioned the precision of the VR anatomical models, pointing out that “some parts looked more like human anatomy than a rat”. Teaching assistants similarly cautioned that students sometimes felt lost without structured objectives, noting the importance of adding “missions or tasks to guide learning”. These observations illustrate the importance of cultivating a critical stance toward the adoption of educational technologies. Students must not only reflect on issues of usability and alignment but also develop the capacity to ask broader ethical questions: Who controls the data generated by these platforms? Who benefits from its collection? Who might be excluded or harmed by its use? Embedding such reflections within immersive learning practices would contribute to the development of responsible digital literacy alongside disciplinary expertise [Zhang et al., 2021].

In addition, the project underscored the importance of collaborative learning and higher-order thinking. Senior students who participated in creating 3D anatomical models reported that the process deepened their own understanding of organ structures. Their contributions also highlighted the role of students as partners in pedagogy, supporting peers while simultaneously building technical and academic expertise. This collaboration fostered a sense of community and inclusivity within the classroom. Moreover, by engaging in content creation, these students moved into higher levels of Bloom’s taxonomy—beyond recall and comprehension toward applying, analyzing, and synthesizing knowledge [Mallek

et al., 2024]. As one student expressed, “Building the model made me think about anatomy in a way I never did before—it’s not just memorizing, it’s really constructing the knowledge”.

7 Conclusion

The HKUST VR Pedagogy Project demonstrates the practical value of immersive technologies in STEM education. By addressing logistical, ethical, and psychological challenges in anatomy teaching, the VR Rat Dissection module provided students with a safer, more engaging, and effective preparation for wet lab experiences.

Looking ahead, situating VR pedagogy within the framework of three educational dimensions—emotional support, creative expression, and critical thinking—offers a more holistic vision of education. This approach not only enhances subject-specific learning but also cultivates capacities essential for navigating the future of education.

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